

The EV Supply Chain Dilemma

Bosch Logical Thinking & Discussion Course

Discussion

Before starting the simulation, review and discuss these key terms and ideas. Understanding them will help you take part in the meeting more effectively.

Competitiveness – How a company compares to others in its industry. A competitive company offers high-quality products at a good price and can adapt quickly to market changes.

ESG (Environmental, Social, and Governance) – A framework companies use to measure their impact on the planet, how they treat people, and how responsibly they are managed. ESG is becoming a major factor in investment and business decisions.

Ethical sourcing – Buying materials and products from suppliers that respect people, communities, and the environment. It means checking that suppliers do not use unfair labor practices or harm the environment.

Supply chain transparency – The ability to see and understand every step in the supply chain—from raw materials to final product—so the company can confirm that partners follow laws and ethical standards.

Due diligence – The careful investigation a company must do before working with another company. It means checking legal, environmental, and social risks before signing a contract.

CSDDD (Corporate Sustainability Due Diligence Directive) – A new rule from the European Union that will require large companies to check their entire supply chain for environmental and human rights risks. It aims to make global business more responsible and transparent.

Discuss with your classmates:

1. Why do you think ESG and ethical sourcing have become important in global business?
2. What might happen if a company ignores due diligence or hides supply chain problems?
3. How could strong supply chain transparency help Bosch maintain its competitiveness and reputation?

The Situation

Bosch Japan is preparing to launch a new line of high-efficiency electric vehicle (EV) components for both domestic and international markets. The project is part of Bosch's global initiative to expand its role in sustainable mobility and reduce its carbon footprint. However, the production of one key component — a lithium-ion battery module — faces a serious supply chain challenge.

A major European supplier has recently increased prices by 30% due to higher energy and labor costs. Meanwhile, a new supplier based in Southeast Asia has offered to supply the same component at a much lower price. On paper, this alternative supplier meets technical specifications and delivery schedules. However, early research suggests that the supplier's environmental and labor practices may not meet Bosch's ESG (Environmental, Social, and Governance) standards. Independent auditors have raised questions about factory emissions, worker safety, and transparency in sourcing materials.

Senior management has called an emergency meeting to decide whether Bosch should:

1. Continue with the current European supplier, despite the high cost;
2. Switch to the Southeast Asian supplier, gaining a strong cost advantage but potential ethical and brand risks; or
3. Adopt a temporary hybrid solution, dividing production between the two suppliers while gathering more data.

Your task in this meeting is to:

1. Discuss the problems and brainstorm options to solve them,
2. Evaluate the risks and benefits of each option, and
3. Decide the best course of action.

The decision will affect production schedules, pricing strategy, brand reputation, and Bosch's long-term ESG commitments.

Note

All facts, figures, and sources in this simulation are fictionalized for educational purposes. They are designed to resemble realistic business data and to help students practice assessing credibility, reliability, and logical reasoning.

Role A: Managing Director (Chair)

You are chairing the meeting and must guide the discussion toward a decision today. Your main goal is to ensure Bosch maintains profitability while protecting its brand image. You are under pressure from headquarters in Germany to reduce production costs by at least 10% by Q4, but you are also responsible for upholding Bosch's sustainability commitments.

Global media attention on corporate ESG compliance has been increasing — in 2023, 78% of Japanese consumers said they would stop buying from a company linked to unethical suppliers (Nikkei survey).

Evidence you can use: Bosch's 2024 Annual Report highlights a 9% profit margin target for its EV division. Any supply change that lowers production costs by more than 5% could significantly support this goal.

Personal view: You prefer to make decisions based on data and risk analysis. You are concerned about long-term reputation and compliance, but you must also meet financial target

Role B: Purchasing Manager

Your role is to manage supplier contracts and cost efficiency. You have verified that the Southeast Asian supplier can meet all technical and logistical requirements and deliver on time. The supplier's price per battery module is ¥27,000, compared to ¥40,000 from the European partner — a cost reduction of nearly 33%. The cost savings could reach ¥1.2 billion per year across all production lines.

You argue that Bosch cannot stay competitive in the EV market if it continues to pay European prices,

especially with strong competition from China and Korea.

Evidence you can use: Bloomberg data shows lithium-ion battery prices have fallen 14% globally in 2024, yet Bosch's current supplier has raised costs.

Additional info: You believe ESG risks can be managed through regular audits and clear supplier contracts with penalties for violations. You have prepared a draft 12-month monitoring plan.

Role C: Sustainability Officer

You are responsible for ensuring all operations comply with Bosch's global ESG policy. You are alarmed by reports from an NGO indicating that the Southeast Asian supplier's factory emits 30% more CO₂ per unit than industry standards and that its subcontractors have been cited for poor worker safety.

Bosch's global sustainability report is due in three months, and any negative findings could have serious reputational consequences.

Evidence you can use: Bosch has pledged to achieve carbon neutrality (Scope 1 & 2) by 2030 and a 15% reduction in Scope 3 emissions by 2027. Partnering with a non-compliant supplier could put these targets at risk.

Additional info: You believe customers and investors increasingly prioritize ethical production — a 2023 Deloitte survey showed that 65% of global investors consider ESG performance a key factor in long-term company value.

Role D: Legal Counsel

Your responsibility is to evaluate the legal and compliance risks. If Bosch knowingly contracts a supplier that violates labor or environmental regulations, it could face fines, lawsuits, and international trade restrictions.

You also know that the EU Corporate Sustainability Due Diligence Directive (CSDDD) will come into effect in 2026, requiring full supply chain transparency for all companies operating in the EU.

Evidence you can use: Non-compliance penalties under similar EU laws can reach up to 5% of global annual turnover. The Southeast Asian supplier's emissions reports lack third-party verification, which would be required under these regulations.

Additional info: If Bosch signs a contract now and problems arise later, the company may be held liable for failing to perform adequate due diligence, even if violations occur abroad.

Role E: Customer Sales Manager

You manage relationships with Bosch's main automotive OEM customers in Japan. If Bosch changes the battery supplier, you must coordinate with customers, since this could affect their production and quality control. You often balance internal cost goals with customer expectations.

Main concerns: Maintaining customer trust, preventing production delays, and giving clear explanations for any supplier change.

Evidence you can use: Japanese OEM customers increasingly value ethical sourcing—72% of automotive buyers say ESG factors influence their decisions (Dentsu Research, 2024). But they also demand stability and consistent quality.

Additional info: Customers will likely need formal documentation and testing approval before accepting a new supplier.

Personal view: The supplier must cut costs while maintaining Bosch's reliability and customer confidence. You prefer a cautious approach that protects long-term relationships.

Role F: Head of R&D (Technical Expert)

You are responsible for product quality and innovation. Both suppliers meet Bosch's basic engineering standards, but the European supplier has a 12-year partnership record with Bosch and a 99.2% on-time delivery rate.

Their failure rate per batch is below 0.3%, compared to the Southeast Asian supplier's trial rate of 2.1%.

Evidence you can use: Delays or quality issues could add ¥300 million in warranty and recall costs if products fail in the market.

Additional info: You also know that the European supplier is testing a new recycling process that could reduce production emissions by 10% by 2026 — aligning closely with Bosch's carbon goals.

Evidence Summary Sheet

Use this sheet after the simulation to review factual evidence, evaluate how it was applied in discussion, and connect data points to the final decision.

Category	Fact / Statistic	Source or Context	Held by before meeting
Profit target	Bosch's 2024 Annual Report shows a 9% profit margin goal for its EV division.	Bosch Corporate Report 2024	Role A: Managing Director (Chair)
Cost comparison	European supplier: ¥40,000 per unit; Southeast Asian supplier: ¥27,000 per unit (33% cheaper).	Purchasing Department Estimates	Role B: Purchasing Manager
Annual savings	Potential ¥1.2 billion cost reduction per year if switching supplier.	Purchasing Manager data	Role B: Purchasing Manager
Global battery prices	Lithium-ion battery prices fell 14% globally in 2024.	Bloomberg 2024 Energy Market Data	Role B: Purchasing Manager
Consumer ethics	78% of Japanese consumers would stop buying from unethical companies.	Nikkei Survey 2023	Role A: Managing Director (Chair)
ESG investor trends	65% of global investors prioritize ESG performance.	Deloitte ESG Survey 2023	Role C: Sustainability Officer
Emissions concern	Southeast Asian supplier emits 30% more CO ₂ per unit than standard.	NGO Environmental Report 2024	Role C: Sustainability Officer
Bosch sustainability goals	Bosch aims for carbon neutrality (Scope 1 & 2) by 2030 and 15% Scope 3 reduction by 2027.	Bosch Sustainability Policy	Role C: Sustainability Officer
Legal risk	EU CSDDD to require full supply chain transparency by 2026.	European Commission Directive Summary	Role D: Legal Counsel
Legal penalties	Fines up to 5% of global annual turnover for ESG violations.	EU Compliance Guidelines	Role D: Legal Counsel
Consumer behavior	72% of Japanese consumers prefer ethical brands; 15% boost in brand favorability among Gen Z engineers.	Dentsu Research 2024 / Bosch HR Report	Mixed provenance: current Role E Customer Sales Manager covers 72% Dentsu/ESG buyer evidence; previous Role E PR & Communications carried 15% Gen Z/HR evidence.
Market competition	Nissan and Panasonic have reduced EV battery costs by 20–25%.	Nikkei Industry Analysis 2024	Previous Role E PR & Communications Manager; no current role-sheet holder found.
Quality control	European supplier failure rate 0.3%, Southeast Asian trial 2.1%.	R&D Quality Audit 2024	Role F: Head of R&D (Technical Expert)
Production risk	Delays or failures could cost ¥300 million in warranty/recalls.	R&D Risk Report	Role F: Head of R&D (Technical Expert)
Innovation opportunity	European supplier's new recycling process may reduce emissions by 10% by 2026.	R&D Collaboration Memo	Role F: Head of R&D (Technical Expert)

Cross-check note: Rows marked in yellow come from a version mismatch. The current Role E is Customer Sales Manager; the previous Role E was PR & Communications Manager. The 15% Gen Z/HR and Nissan/Panasonic competition evidence came from the previous Role E and was not reassigned in the current role sheets.

Follow-Up

After the simulation, discuss:

- Which argument was the most persuasive? Why?
- How well did participants use reasons and evidence to support their points?
- What would you do differently next time to improve logical structure and communication clarity?